

# Summary - 2403.06535V1 Copy

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## 1. Overall Summary

This paper introduces DeLAMA, a novel decentralized and lifelong-adaptive multi-agent collaborative learning algorithm designed to enhance collaboration among multiple agents without a central server, even when agents face varied and evolving tasks. The core problem addressed is the limitations of existing multi-agent collaboration methods, which often rely on predefined, static collaboration relationships or centralized servers, leading to vulnerability and inefficiency in dynamic environments. DeLAMA tackles this by proposing a decentralized graph structure learning algorithm for autonomous collaboration relationship discovery and a memory unit for lifelong adaptation to dynamic tasks, ensuring finite storage consumption. The system integrates mathematical optimization with algorithm unrolling, transforming iterative solutions into neural network layers with learnable parameters, allowing agents to 'learn to collaborate' through supervised training. Theoretical analysis confirms communication efficiency and convergence properties. Experimental results on synthetic and real-world datasets (regression, classification, multi-robot mapping) demonstrate DeLAMA's superior performance, achieving a 98.80% reduction in MSE and a 188.87% improvement in classification accuracy compared to federated learning and other decentralized methods. The system effectively discovers collaboration strategies, adapts to dynamic scenarios, and shows compatibility with human cognitive collaboration mechanisms, paving the way for intelligent, decentralized, and dynamic multi-agent systems.

## 2. Main Idea

The central concept is DeLAMA, a novel decentralized and lifelong-adaptive multi-agent collaborative learning algorithm. It enables multiple agents to autonomously identify beneficial collaboration relationships and adapt to dynamically changing tasks over time, without relying on a central server or predefined static collaboration structures. This is achieved by combining mathematical optimization with algorithm unrolling techniques to leverage neural networks for enhanced learning ability, flexibility, and computational efficiency.

### 3. Problem Being Solved

This paper addresses several weaknesses and research gaps in previous multi-agent collaborative learning systems: 1. **Reliance on predefined, static collaboration relationships:** Existing methods often use fixed collaboration graphs (e.g., fully connected, star graphs) or task correlations calculated by a central server, which are inadequate and inflexible for dynamic scenarios where agent observations and tasks evolve over time. 2. **Vulnerability of centralized collaboration mechanisms:** Collaboration guided by a central server (e.g., federated learning) creates a single point of failure, disabling all collaborations if the central server fails. 3. **Lack of autonomous collaboration discovery:** Agents in previous systems often cannot autonomously identify beneficial collaborators, leading to redundant or low-quality information exchange. 4. **Inefficient adaptation to dynamic tasks:** Static collaboration models limit the system's ability to adapt efficiently to continuously changing observations and tasks, hindering overall collaboration effectiveness and learning performance over time.

### 4. Assumptions

- The training data arrives independently among the agents at each time  $t$ , and the data batches  $D_i(t)$  satisfy conditional independency on both agent-level for  $1 \leq i \leq N$  and time-level for  $1 \leq k \leq t$ .
- Similar model parameters promote agents to collaborate, where more similar model parameters correspond to stronger collaboration relations.
- The communication graph  $C(t)$  is known.
- The adjacency matrix  $W(t)$  is restricted by the communication graph  $C(t)$  or its substructure.
- The storage space for the learning memory  $M_i(t)$  of the entire learning task  $T_i$  is finite, prohibiting direct storage of past training data.
- The model parameters initialized by local learning are  $\theta_i(t,0) = [A_i(t) + 2\lambda_{-1}I]^{-1} b_i(t)$ .
- The adjacency matrix  $W(t)$  is symmetric.

### 5. Limitations

- The Taylor expansion used in the local learning step ( $\Phi_{\text{local}}(\cdot)$ ) has potential approximation errors for non-linear functions, limiting the method's expressive capability for non-linear learning tasks.
- The iterative algorithm involves many hyperparameters, demanding significant effort for searching and tuning.
- The algorithm requires numerous iterations (single graph learning, message passing, and alternating convex search), leading to significant communication overhead and making it less practical for real applications.

## 6. Results and Conclusion

The experimental findings demonstrate DeLAMA's superior performance and capabilities: \* \*\*Regression and Classification:\*\* DeLAMA significantly outperforms federated learning and decentralized optimization methods, achieving a 98.80% reduction in Mean Squared Error (MSE) and a 188.87% improvement in classification accuracy at later timestamps ( $t=10$ ). \* \*\*Collaboration Graph Learning:\*\* It effectively learns collaboration relationships under various communication structure constraints (fully connected, Erdos-Renyi, Barabasi Albert graphs), outperforming traditional centralized and decentralized structural learning methods, and matching the performance of recent structural learning approaches like L2G. \* \*\*Lifelong Adaptation:\*\* DeLAMA exhibits strong lifelong-adaptive capabilities, efficiently remembering previously learned data and adapting to new knowledge, significantly outperforming classic lifelong learning approaches (e.g., LWF, EWC, MAS, GEM, A-GEM) by up to 99.66% on MNIST and 134.01% on CIFAR-10 at  $t=10$ . \* \*\*Dynamic Evolution:\*\* The system dynamically evolves collaboration relationships over time, initially collaborating broadly and then refining connections to form accurate groups as observations grow, aligning with ground truth structures. \* \*\*Multi-Robot Mapping:\*\* In multi-robot 2D mapping tasks, DeLAMA outperforms other decentralized collaborative model training approaches (DSGD, DSGT, DiNNO) in F1-score, demonstrating improved environmental exploration and more complete map structures through collaboration. \* \*\*Human-Involved Experiment:\*\* A human experiment confirms that DeLAMA's collaboration mechanism is compatible with human cognitive collaboration, showing a negative correlation between human rating scores (collaboration strength) and model parameter distances, verifying the assumption that humans tend to collaborate with similar knowledge. In conclusion, DeLAMA successfully provides a robust, communication-efficient, and adaptive framework for multi-agent collaborative learning in dynamic, decentralized environments.

## 7. Real World Impact

This research has practical real-world applications in various domains: \* \*\*Autonomous Driving:\*\* Vehicle-to-everything (V2X) communication-aided autonomous driving, where vehicles can collaboratively perceive and make decisions. \* \*\*Robotics:\*\* Multi-robot environment exploration and multi-robot mapping, enabling robots to efficiently map unknown areas through collaboration. \* \*\*Edge AI/Distributed Machine Learning:\*\* Collaborative training of machine learning models across multiple clients, which is crucial for edge AI and distributed computing scenarios, potentially in smart healthcare or industrial IoT. \* \*\*Smart Infrastructure:\*\* Traffic control and unmanned aerial vehicle (UAV) system control, where agents (e.g., traffic lights, drones) can coordinate to optimize system performance.

## 8. Graphs and Figures Analysis

**Figure 1: The illustration of our decentralized and lifelong-adaptive multi-agent collaborative learning system.**

This figure visually represents the DeLAMA system. It shows six agents, each with model parameters, observations, and collaboration relationships. The key aspect is that these elements (observations, tasks, and thus collaboration relationships) are dynamic and evolve over time (Timestamp 1, 2, 3). The agents autonomously adapt their collaboration relationships to these changing tasks without a central server.

**Figure 2: Relationships among our lifelong collaborative learning and its related works: lifelong learning, federated learning, and multi-task relation learning.**

This 2x2 matrix plot categorizes different learning paradigms based on two axes: 'Dynamic scenario' (top row) vs. 'Static scenario' (bottom row) and 'Predefined collaboration relations' (left column) vs. 'Learned collaboration relations' (right column). The top-left shows Lifelong learning (dynamic, predefined relations), bottom-left shows Federated learning (static, predefined relations), bottom-right shows Multi-task relation learning (static, learned relations), and top-right shows Decentralized and lifelong-adaptive collaborative learning (DeLAMA) (dynamic, learned relations). This highlights DeLAMA's unique position in addressing both dynamic scenarios and learned collaboration.

**Figure 3: The decentralized and lifelong-adaptive multi-agent collaborative learning system.**

This figure illustrates the three main steps of the DeLAMA system: 1) Local learning, where each agent learns model parameters based on its own observations for initialization. 2) Collaboration relational inference, where agents transmit model parameters along the communication graph and learn collaboration relations ( $W(t)$ ). 3) Lifelong model update, where agents share model parameters along the learned collaboration relation with collaborators and refine their parameters. The second and third steps are iterated until convergence. The figure also shows 'Agents with different observation,' 'Communication graph constraint,' and 'Collaboration relation' as key components.

**Figure 4: The unrolled network structure of the collaboration system with input data at one exact time.**

This figure depicts how the iterative optimization steps are unrolled into a deep neural network. The 'Agent' (at Time  $t$ ) processes input data ( $D_i(t)$ ) through a backbone network ( $\Phi_\beta$ ) to calculate Hessian ( $H_i(t)$ ) and update history ( $A_i(t)$ ,  $b_i(t)$ ), leading to initialized model parameters ( $\theta_i(t,0)$ ). 'Layer 1' and 'Layer M1' ( $\Phi_{\text{graph}}$ ) show agents communicating to infer collaboration relationships ( $W^*(t)$ ). 'Layer 1' and 'Layer M2' ( $\Phi_{\text{param}}$ ) show agents aggregating model parameters from collaborators and refining their own parameters ( $\theta_i(t,k)$ ). The final output is the refined model parameters ( $\theta^*(t)$ ), which are supervised by training tasks. This unrolling allows learning to tune hyperparameters ( $\gamma$ ) for optimal strategies.

**Figure 5: Visualization of the learned graph structure at time  $t = 3$  compared with the oracle collaboration structure.**

This figure presents heatmaps visualizing the learned collaboration graph structures at time  $t=3$  for different methods (GL-LogDet, GLasso, MTRL, L2G-PDS, NS, DeLAMA) compared to the Oracle structure. Each heatmap shows a  $6 \times 6$  matrix where cell intensity indicates collaboration strength between agents. DeLAMA's heatmap closely resembles the Oracle structure, indicating it learns smooth and accurate edge weights, outperforming other baseline methods which show less accurate or sparser collaboration patterns.

**Figure 6: A collaboration result of two groups of agents that are performing two different regression tasks.**

This figure shows two parts: (a) 'Functions learned by each agent' plots learned regression functions for six agents over five timestamps ( $T=1$  to  $T=5$ ). Agents are divided into two groups, each performing a different regression task. The plots show that as time progresses, agents refine their learned functions, indicating adaptation and knowledge accumulation. (b) 'Time evolving collaboration structure' displays heatmaps of the collaboration graph structure for the six agents over five timestamps. Initially, agents might have broader connections, but as time evolves, the collaboration structure refines into two distinct groups, reflecting the two different tasks being performed. This demonstrates DeLAMA's ability to dynamically evolve collaboration relationships.

**Figure 7: Example of the evolving collaboration graphs.**

This figure displays heatmaps of evolving collaboration graphs for MNIST (a) and CIFAR-10 (b) datasets at different timestamps ( $T=1, T=3, T=5, T=7$ ) compared to the Oracle (ground truth) map. The heatmaps show that initially, agents might collaborate more broadly ( $T=1$ ), but as time progresses and observations grow, the system tends to identify and cut off less useful relationships, forming more refined and accurate collaboration groups that align with the Oracle structure. This illustrates DeLAMA's dynamic adaptation of collaboration relationships.

**Figure 8: The simulation result of robot mapping at  $t = 12$  based on the CubiCasa5k room Dataset.**

This figure presents visual results of a multi-robot 2D mapping task at time  $t=12$  for different decentralized collaborative model training approaches (DeLAMA, DeLAMA-WC, DSGD, DSGT, DINNO) compared to a 'Truth Map.' Black represents empty space, and white represents occupied space. DeLAMA's map (second from left) most closely resembles the 'Truth Map,' indicating superior performance in accurately mapping the environment through collaboration, compared to other methods which show more incomplete or erroneous maps. DeLAMA-WC (without collaboration) shows a significantly less complete map, highlighting the benefit of collaboration.

**Figure 9: The human evaluation of collaboration relationships with each other.**

This figure shows three scatter plots, each representing a different data quality (High, Medium, Low). The x-axis represents human rating scores (0 to 5, higher means higher collaboration), and the y-axis represents the logarithm of model parameter distance ( $\log(D_{ij})$ ). The plots show a negative correlation: as human collaboration scores increase (meaning stronger collaboration), the model parameter distance between agents tends to decrease. This verifies the assumption that humans tend to collaborate with similar agents. For lower data quality, the correlation appears less clear, suggesting humans have more difficulty identifying useful collaborators in harder problems.

**Figure 10: A time-evolving learning task example of two agents sampled from the regression task dataset.**

This figure shows the learned regression functions for two agents over five timestamps ( $T=1$  to  $T=5$ ), along with the ground truth function. Each agent initially has a limited sampling interval, and their learned function is far from the ground truth. As time progresses ( $T=1$  to  $T=5$ ), the agents' knowledge of the target function improves, and their learned regression lines get closer to the ground truth, demonstrating the lifelong learning ability and adaptation to increasing sampling intervals.

**Figure 11: The human-machine interaction framework used for the human-involved experiment.**

This diagram illustrates the framework for the human-involved experiment. It shows a 'Human machine interaction' layer where humans visually interact and imitate, and a 'Background algorithms' layer where DeLAMA and 'Proxy machines' (imitating human behaviors) operate. The interaction involves humans observing virtual agents' behaviors and submitting their understandings, with virtual agents performing visual interaction. This top-down framework facilitates human-machine communication and evaluation.

**Figure 12: The login page of the human-involved experiment, contains a user guide for the participants to learn how to play the game.**

This figure displays a screenshot of the login page for the human-involved experiment. It includes fields for User ID and Username, and a 'How to play the game?' section providing instructions for participants on the linear regression task, emphasizing that they only access a portion of data and can modify their answers based on other players' results. Small example plots for Time 1, 2, 20 are shown.

**Figure 13: The local learning page of the human-involved experiment.**

This figure shows a screenshot of the local learning page from the human-involved experiment. It displays a scatter plot of data points and allows the participant to draw a line (representing their guess of the regression line) by inputting an equation ( $y = 0.39x + -4.60$ ). This page represents the initial learning phase based purely on visual information.

**Figure 14: The aggregation page of the human-involved experiment, which contains other participants' results and the score rated by each agent.**

This figure shows a screenshot of the aggregation page. It displays the participant's current regression line along with the regression lines of other participants. It also shows an 'Evaluation of your performance' score and a rating mechanism (stars) for other agents, allowing participants to see and potentially modify their results based on collaborators' performance.

**Figure 15: The final page of the human-involved experiment with a leaderboard, each participant can look back to historical learning results.**

This figure shows a screenshot of the final page of the human-involved experiment. It includes a 'Leaderboard' displaying average performance scores and a 'Result Log' showing the evolution of the participant's evaluation loss over different timestamps ( $T=1$  to  $T=4$ ), along with the ground truth regression target function. This allows participants to review their historical learning progress and compare it with others.